Spam Mail Detector

Objective

The objective of this project is to build a machine learning model that can classify SMS messages as either Spam or Ham (Not Spam).

Dataset

SMS Spam Collection Dataset

The dataset contains SMS messages labeled as:

- Spam
- Ham (Not Spam)

Technologies Used

- Python
- Pandas
- Scikit-Learn
- TF-IDF Vectorization
- Naive Bayes Classifier

Workflow

1. Load the SMS dataset.
2. Preprocess text data.
3. Convert text into numerical features using TF-IDF.
4. Split data into training and testing sets.
5. Train a Multinomial Naive Bayes classifier.
6. Evaluate model performance.
7. Predict whether a new message is spam or not.

Machine Learning Algorithm

Multinomial Naive Bayes

Naive Bayes is a probabilistic classification algorithm widely used for text classification problems such as spam detection.

Evaluation Metrics

- Accuracy Score
- Precision
- Recall
- F1-Score

Results

The model successfully classifies SMS messages into spam and ham categories with high accuracy.

Sample Prediction

Input:

Congratulations! You won a free iPhone. Click here.

Output:

Spam Message

Skills Learned

- Natural Language Processing (NLP)
- Text Preprocessing
- TF-IDF Feature Extraction
- Classification Models
- Model Evaluation

Author

Sravya